

An Autonomous Agent that Derives ADC Linker Rules from the Literature, Designs the Molecules, and Flags When the Field Disagrees

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Abstract

The right linker for an antibody conjugate depends on its payload: cytotoxins favour cleavable release, immune-stimulating conjugates demand plasma stability, and antibody–oligonucleotide conjugates were reported to need a rigid non-cleavable linker. We built an agent that reads 31 primary papers, derives a design rule per payload class with a retrieval-grounded language model, and lets each rule parameterise a REINVENT4 objective that then generates the linkers, so the delivered molecules carry the class motif by construction. A leave-one-paper-out test shows the cytotoxin and ISAC rules are robust—withhold a paper and the agent re-derives the same cleavable rule (confidence 0.97, 0.83)—while the ARC rule is contested: withholding the rigid-non-cleavable siRNA paper makes the agent derive the opposite rule, protease-cleavable Val-Cit, from the clinical antibody-oligonucleotide literature, and its confidence falls to 0.33 because the grounded evidence genuinely splits. The rule-generated Val-Cit design is recognised by cathepsin B (predicted K_d 9 nM) as tightly as the clinical substrate, while the rigid ARC design is not. Every reported value is provenance-tagged (a real model run, an LLM completion, or a cheminformatics heuristic) and a gate blocks any placeholder from a result—so the contested-ARC finding is a grounded, calibrated conclusion a lookup-table system could not produce.

1 Introduction

Antibody conjugates couple antibody selectivity to a potent cargo, and the linker sets their therapeutic index [1, 3]. The linker requirement inverts with payload class: cytotoxins favour cleavable release [2], ISACs demand plasma stability [6, 7], and antibody–oligonucleotide conjugates (ARCs) were reported to favour rigid non-cleavable linkers [5]. An autonomous scientist should derive such rules from the literature, use them to design molecules, and report how confident it is—including recognising where the literature genuinely conflicts. We show all three, and the last surfaces a real, unresolved tension in ARC linker design (Figure 1).

2 Methods

2.1 Reasoning that drives generation

For each payload class an agent (Claude Opus, always on) queries a store of 31 papers, retrieves passages with provenance, and extracts conjugate exemplars that must each cite a retrieved passage. It derives a rule (cleavage, rigidity, plasma-stability) from the grounded exemplars—with no hardcoded fallback, so a failed call is an explicit abstain—and the rule compiles into an ADC goal profile (cleavage regime, rotatable-bond window, per-term weights) that parameterises the REINVENT4 LinkInvent objective. That objective then generates the linkers (staged learning, remote GPU); the class trigger warhead (Val-Cit-PABC, Val-Ala-PABC or the rigid sulfo-SMCC cap) is welded into every molecule (Figure 3).

2.2 Confidence, scorers and provenance

Rule confidence is *disagreement-aware*: it is high only when the grounded exemplars are both plentiful and consistent, so a split field drives confidence down and is flagged contested. Synthesizability is the continuous Ertl SA_Score (the metric the generator optimises); the step-count survives only as a reported complexity index, not a route. Plasma stability is mechanism-resolved. Every value that reaches the manuscript is tagged *measured* (a real Boltz or REINVENT run), *llm*, or *heuristic*; a gate blocks any placeholder from a results claim (SI provenance table).

One molecule, end to end: a retrieved quote grounds the rule that generates the linker that cathepsin B recognises

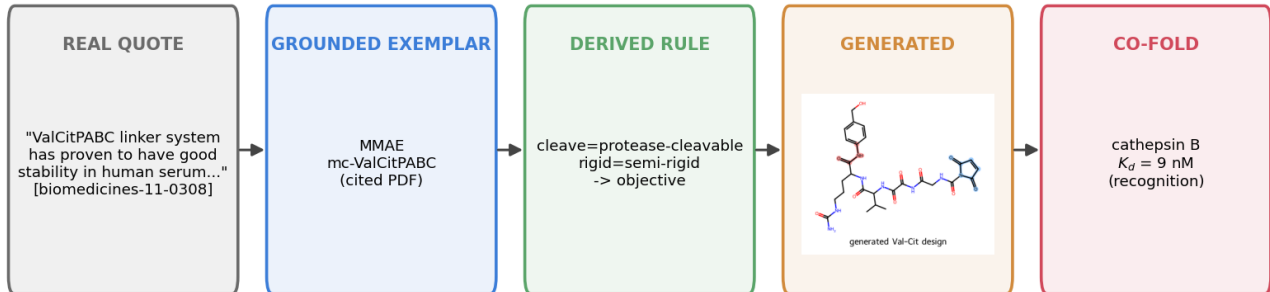


Figure 1: One molecule, end to end: a retrieved quote grounds the cytotoxin rule that generates a Val-Cit linker that cathepsin B recognises. Handle (blue), scissile bond (red) and spacer (green) are SMARTS-detected.

Table 1: Leave-one-paper-out. Confidence is disagreement-aware; “evidence” is the grounded cleavable/non-cleavable split. The ARC rule flips direction and loses confidence because its evidence is genuinely divided.

Class	Withheld	Full-corpus	Held-out	Evidence	Outcome
Cytotoxin	Balamkundu 2023	cleavable (0.60)	cleavable (0.97)	8/0	recovers
Immunomodulator (ISAC)	imidazoquinoline ISAC	cleavable (0.83)	cleavable (0.83)	1/0	recovers
Oligonucleotide (ARC)	ARC siRNA (Bioconj. Chem. 2025)	non-cleavable (0.50)	cleavable (0.33)	4/2	flips, contested

2.3 Held-out test and co-fold

We withhold a class’s key paper, re-derive the rule from the remaining corpus, record the prediction before revealing the withheld paper, and report agree/disagree with its evidence composition. We co-fold each designed linker with its real payload against cathepsin B (Boltz-2) as two co-present ligands—a recognition/plausibility check, not a covalent conjugate and not a claim of cleavage.

3 Results

3.1 The rules generate, and steer, the molecules

Each derived rule compiles into a distinct objective (Figure 3, left) that generates its linkers: the designs separate cleanly by class along the two rule knobs (Figure 3, right), ARC in a rigid non-cleavable region (4–6 rotatable bonds), cytotoxin and ISAC as cleavable flexible peptides (12–16). The molecules are made by the reasoning, not labelled after the fact.

3.2 Held-out test: robust rules and a contested one

The leave-one-paper-out test discriminates (Figure 2, Table 1). The cytotoxin and ISAC rules are robust: their grounded exemplars are unanimously cleavable, so withholding a paper leaves the derived rule and its high confidence intact. The ARC rule is contested. With the full corpus the top-retrieved siRNA papers give rigid non-cleavable. Withhold them, and from the clinical antibody-oligonucleotide literature (DYNE-101/251, AOC-1001) the agent derives protease-cleavable Val-Cit instead—its direction flips, and because the grounded evidence splits four cleavable to two non-cleavable, its disagreement-aware confidence falls to 0.33. Early cationic-assistance-free siRNA conjugates favour rigid non-cleavable linkers; clinical AOCs increasingly use cleavable Val-Cit. The agent surfaces that tension and lowers its confidence accordingly, rather than asserting a single rule.

3.3 Structural recognition

The rule-generated Val-Cit cytotoxin design co-folds tightly with cathepsin B (predicted K_d 9 nM), near the clinical mc-Val-Cit-PABC substrate (35 nM), while the rigid non-cleavable ARC design binds an order of magnitude more weakly (109 nM) (Figure 4). Affinity indexes recognition, not cleavage.

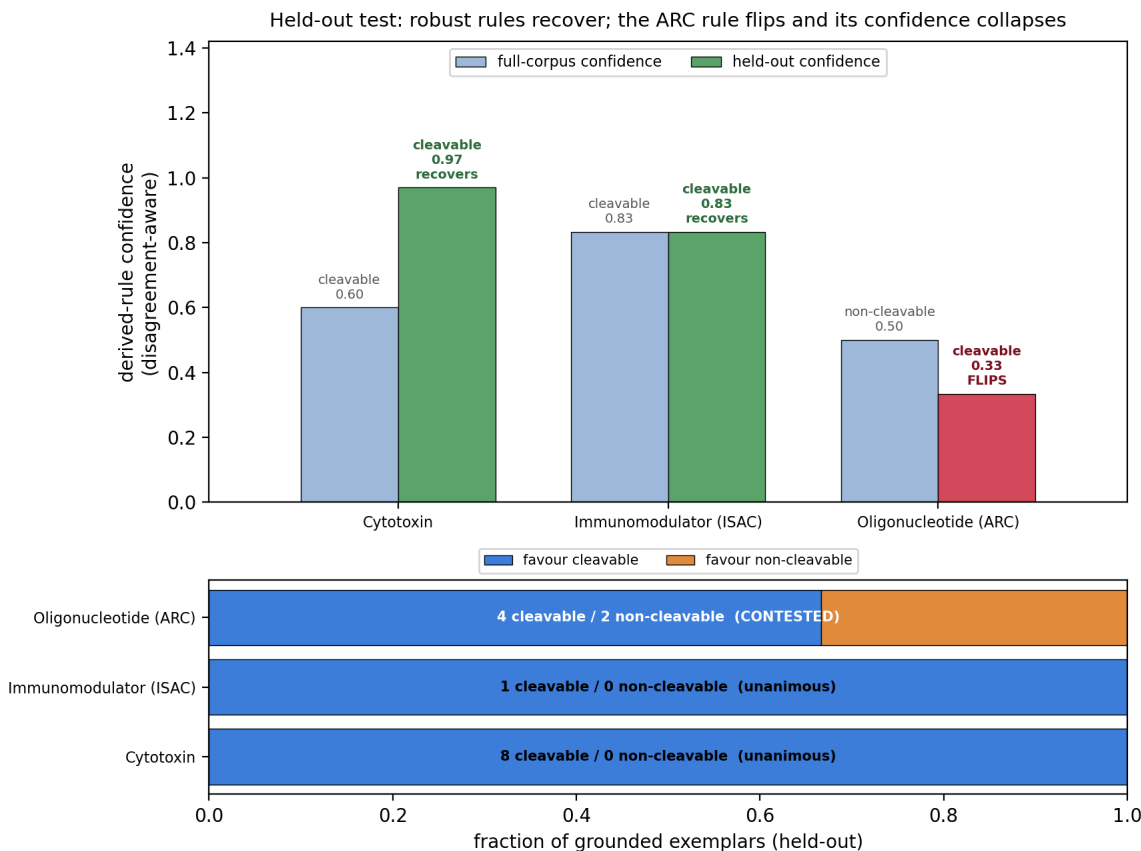


Figure 2: Held-out test. Top: full-corpus vs held-out confidence (disagreement-aware); the ARC rule flips (red) while cytotoxin/ISAC recover (green). Bottom: the grounded-evidence split that explains it—ARC is 4:2 (contested), the others unanimous.

Table 2: One representative generated design per class (all fifteen with full dossiers in the SI). SA = continuous Ertl synthetic-accessibility subscore; Stab = mechanism-resolved plasma-stability score.

Class	Representative generated linker	SA	Stab	Nearest clinical (Tc)
Cytotoxin	<chem>CC(C)C(NC(=O)C(=O)NC(=O)CNC(=...</chem>	0.6437	0.85	mc-Val-Cit-PABC (0.654)
Immunomodulator (ISAC)	<chem>CC(NC(=O)C(NC(=O)CC(N)C(=O)NC...</chem>	0.6547	0.85	mc-Val-Ala-PABC (0.58)
Oligonucleotide (ARC)	<chem>NCC1CCC(NC(=O)N2CCN(c3ccc(S=...</chem>	0.6866	1.0	MCC (non-cleavable (0.145))

3.4 Actionable designs

Table 2 gives one representative generated design per class (all fifteen, with dossiers—route, cost, failure modes, validation—in the SI). Structures are drawn with SMARTS-detected handle/scissile/spacer highlights (Figure 1).

4 Discussion

The agent reads the literature, derives rules that generate synthesisable molecules, and reports calibrated confidence that falls when the evidence divides—turning “contested” from a caption into a computed output. The ARC result is the payoff: a grounded, non-obvious finding that clinical antibody-oligonucleotide conjugates favour cleavable Val-Cit, against the rigid-non-cleavable rule of the early siRNA literature. *Limitations.* The complexity index is a count-based proxy, not a retrosynthetic route (AiZynthFinder is the planned upgrade); the co-fold places linker and payload as co-present ligands, not a covalent construct; the retrieval is lexical; and generation samples 25 designs per class. Each is tagged, not hidden.

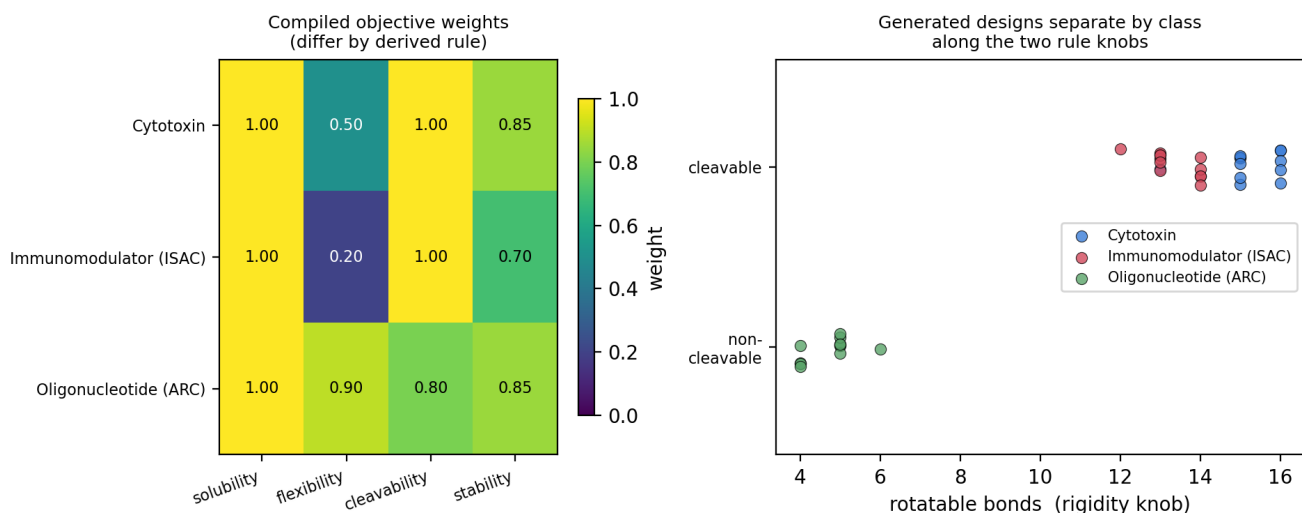


Figure 3: The rules steer the chemistry. Left: the compiled objective weights differ by class. Right: the generated designs separate by class along the two rule knobs (rotatable bonds; cleavable-motif presence).

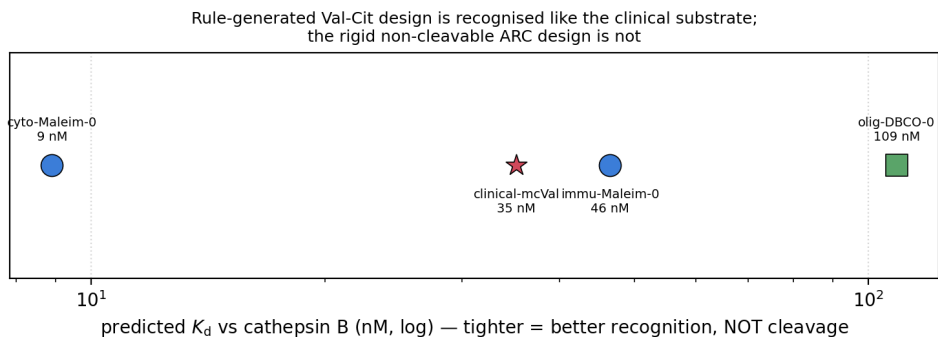


Figure 4: Cathepsin-B co-fold. The rule-generated Val-Cit design binds as tightly as the clinical substrate; the rigid non-cleavable ARC design does not. Predicted affinity indexes recognition, not cleavage.

5 Conclusion

An autonomous agent derived payload-class ADC-linker rules from 31 papers, used them to generate synthesisable linkers, and, under a held-out test, distinguished robust rules from a genuinely contested one—surfacing an unresolved question in ARC linker design and lowering its confidence where the literature divides. The approach points toward an autonomous medicinal-chemistry scientist that designs and knows what it does not know.

Acknowledgements

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